

Lunar Surface Infrastructure Program Charter

Revised and restated educational planning baseline version 1.1

Revision markers identify material added or expanded from version 1.0.

Sponsor	Self-directed educational initiative	Status	Planning baseline version 1.1
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Authorization and purpose

Authorize concept planning for an integrated lunar surface demonstration that uses robotic systems to survey, prepare, and verify a selected site; establishes essential systems for a seven-person crew; and commissions a resilient lunar data center without compromising crew safety. The charter establishes direction, boundaries, governance, measures, and decision gates. It does not represent an approved flight mission or validated engineering design.

Mission objective Demonstrate that robotic deployment, human habitation, shared infrastructure, and computing operations can be integrated and controlled within defined safety, schedule, resource, and performance limits.	Planning assumptions <i>expanded in v1.1</i> Seven crew members; 45-day nominal surface mission; consumables and critical reserves sized for at least 60 days; illuminated south-polar ridge as the reference site; robotic assets arrive before the crew; essential infrastructure is remotely verified before crew launch.
Program scope <i>expanded in v1.1</i> 1 Program integration and systems engineering 2 Launch and lunar transportation services 3 Robotic deployment and site preparation 4 Seven-person habitat readiness 5 Lunar data-center commissioning and operations 6 Shared infrastructure and resilience	Operating priorities <i>expanded in v1.1</i> 1 Protect crew life and safe departure capability 2 Preserve habitat and critical communications 3 Maintain shared power, navigation, and hazard monitoring 4 Complete robotic site and infrastructure objectives 5 Deliver critical computing 6 Execute discretionary science and computing
Program boundaries <i>clarified in v1.1</i> Educational planning and simulation only. Launch and lunar-delivery services are modeled or procured; no new launch vehicle is developed. Flight-certified design, vehicle selection, contract commitments, medical certification, detailed cost certification, and lunar resource ownership claims remain outside scope.	Program leadership <i>expanded in v1.1</i> Program Director owns the integrated baseline, priorities, resource strategy, risk posture, readiness recommendations, change control, and executive communication. Project leads own delivery. The Chief Engineer owns technical integration. Safety and Mission Assurance retains independent stop-work authority. A Configuration Control Board evaluates baseline changes.

Success measures *expanded in v1.1*

Program integration	Crew and habitat	Robotic deployment	Data center and resilience
Gate-entry criteria met; interfaces and integrated schedule baselined; no unaccepted critical risk.	No loss of critical life-support capability; survival reserves remain above thresholds; safe departure remains available.	Survey, preparation, placement, inspection, and remote verification completed to approved acceptance criteria.	Critical computing operates within protected utility margins; hazards are detected; critical functions recover within approved targets.

Decision gates *G2 added and later gates renumbered in v1.1*

G1 Program baseline	G2 Launch and robotic readiness	G3 Site readiness
Authorize integrated planning using approved objectives, scope, organization, assumptions, WBS approach, schedule framework, and initial risk register.	Authorize robotic mission execution using payload integration, launch and delivery, communications, navigation, landing, verification, and contingency evidence.	Accept the site for habitat deployment using survey results, terrain and landing hazards, preparation evidence, utilities, communications, and monitoring.
G4 Crew launch	G5 Data-center activation	G6 Mission closeout
Authorize crew departure after verifying habitat pressure, life support, power, communications, supplies, shelter, medical support, and departure capability.	Authorize critical computing after confirming stable habitat operations and approved power, thermal, communications, cybersecurity, maintenance, and emergency margins.	Accept mission data, residual risk, asset configuration, lessons learned, and follow-on recommendations.

Operating rule: Habitat survival loads remain protected ahead of computing and discretionary mission loads.

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Version 1.1 execution baseline additions

Initial risks and controls *expanded from 3 to 8 risk categories in v1.1*

Risk	Potential program effect	Initial control approach
Launch, transit, or landing failure	Loss or delay of robotic, habitat, utility, or computing elements disrupts the integrated sequence.	Phased manifests, alternate delivery opportunities, qualification evidence, schedule reserve, and mission-specific abort or disposal planning.
Robotic deployment or autonomy failure	Site remains unverified or critical infrastructure cannot be prepared before crew launch.	Redundant assets, supervised autonomy, remote inspection, safe modes, repair tools, spares, and a no-go crew-launch gate.
Site or terrain uncertainty	Landing, mobility, construction, illumination, or communications performance is worse than assumed.	Progressive reconnaissance, site alternatives, hazard maps, mobility trials, margin, and evidence-based acceptance criteria.
Shared power or thermal shortfall	Competition between habitat and computing reduces survival reserve or mission output.	Protected survival bus, load-shedding hierarchy, independent emergency power, storage margin, heat-rejection margin, and trend reviews.
Life-support or habitat failure	Crew health, mission duration, or safe return is threatened.	Functional redundancy, isolation, emergency oxygen and water, repair capability, storm shelter, health monitoring, and departure criteria.
Natural hazard exposure	Radiation, dust, micrometeoroids, eclipses, or thermal extremes degrade multiple systems.	Shielding, protected placement, dust controls, hazard sensing, predictive maintenance, safe modes, spares, and recovery exercises.
Communications, navigation, or cyber disruption	Remote control, crew coordination, situational awareness, or data integrity is lost.	Diverse links, local autonomy, authenticated commands, segmented networks, store-and-forward capability, and degraded-mode procedures.
Logistics or departure delay	Consumables, spares, or crew exposure exceed the planned mission duration.	Sixty-day provisioning baseline, reserve policy, repair prioritization, contingency schedule, and continuous departure-readiness monitoring.

Authorized planning products *added in v1.1*

- Program Level 1 and Level 2 work breakdown structure with a WBS dictionary and accountable owners.
- Integrated milestone schedule showing robotic arrival, site verification, habitat readiness, crew launch, and data-center activation dependencies.
- Requirements and interface baseline covering launch, landing, robotics, habitat, shared utilities, computing, operations, and safety.
- Responsibility assignment matrix and governance cadence for decisions, change control, reviews, and escalation.
- Initial risk and opportunity register with probability, consequence, owners, triggers, mitigation actions, and retirement criteria.
- Verification and validation strategy with readiness criteria for each program gate.

Immediate next decision

Approve development of the integrated WBS and WBS dictionary using this charter as the scope baseline. The first decomposition will establish accountable work packages across the three projects and shared enabling systems before schedule dates or resource commitments are assigned.

Revision record

Version	Date	Summary of change
1.0	20 September 2026	Established the initial educational planning baseline for robotic deployment, seven-person habitat commissioning, and lunar data-center operations.
1.1	21 September 2026	Expanded transportation scope, operating priorities, governance, success measures, decision gates, program risks, and authorized planning products while retaining the original mission direction.

Reference basis: NASA lunar south-pole candidate-region considerations and the NASA human-spaceflight hazard framework. All mission durations, capacities, thresholds, configurations, and performance figures remain educational planning assumptions pending analysis and verification.